

# Machine Learning Foundation Curriculum

Beginner → Advanced | Focused foundational syllabus

**Goal:** Build a strong, practical Machine Learning foundation through structured progression from fundamentals to practical model development.

## LEVEL 1 — MACHINE LEARNING FUNDAMENTALS

### ***1. Introduction to Machine Learning***

- What is Machine Learning?
- AI vs Machine Learning
- Types of Machine Learning: Supervised, Unsupervised, Reinforcement Learning
- Training, Validation, Testing, Inference
- Features, Labels / Targets, Samples, Dataset
- Model, Parameters, Hyperparameters

### ***2. Machine Learning Workflow***

- Problem definition
- Data collection
- Data preparation
- Data preprocessing
- Feature engineering
- Dataset splitting
- Model selection
- Model training
- Model evaluation
- Hyperparameter tuning
- Prediction

## LEVEL 2 — DATA HANDLING

### ***3. NumPy Fundamentals***

- Arrays
- Dimensions
- Shape
- Indexing
- Slicing
- Reshaping
- Broadcasting
- Vectorized operations
- Matrix operations
- Random numbers

#### **4. Pandas Fundamentals**

- Series
- DataFrames
- Loading data
- Selecting data
- Filtering
- Sorting
- Grouping
- Aggregation
- Merging
- Missing values
- Duplicate values

#### **5. Data Types**

- Numerical data
- Categorical data
- Discrete data
- Continuous data
- Structured data
- Unstructured data

### **LEVEL 3 — DATA PREPROCESSING**

#### **6. Data Cleaning**

- Missing values
- Duplicate values
- Invalid values
- Inconsistent values
- Outliers
- Data type conversion

#### **7. Feature Transformation**

- Encoding
- Label Encoding
- One-Hot Encoding
- Ordinal Encoding
- Feature scaling
- Standardization
- Normalization
- Log transformation

#### **8. Dataset Splitting**

- Training set
- Validation set
- Test set
- Random split
- Stratified split
- Data leakage

### **LEVEL 4 — EXPLORATORY DATA ANALYSIS**

## **9. Descriptive Statistics**

- Mean
- Median
- Mode
- Variance
- Standard deviation
- Percentiles
- Quartiles
- IQR

## **10. Data Visualization**

- Histograms
- Box plots
- Scatter plots
- Bar charts
- Line charts
- Distribution plots
- Correlation heatmaps

## **11. Correlation**

- Covariance
- Correlation
- Pearson correlation
- Spearman correlation
- Feature relationships
- Multicollinearity

# **LEVEL 5 — REGRESSION**

## **12. Linear Regression**

- Simple linear regression
- Multiple linear regression
- Regression equation
- Parameters
- Cost function
- MSE
- Prediction
- Gradient descent

## **13. Polynomial Regression**

- Polynomial features
- Model complexity
- Underfitting
- Overfitting

## **14. Regularization**

- L1 regularization
- L2 regularization
- Ridge
- Lasso
- Elastic Net

# **LEVEL 6 — CLASSIFICATION**

## **15. Logistic Regression**

- Binary classification
- Sigmoid function
- Probability
- Decision boundary
- Log loss
- Multiclass classification

## **16. K-Nearest Neighbors**

- Distance
- Choosing K
- Classification
- Regression

## **17. Decision Trees**

- Tree structure
- Nodes
- Splitting
- Entropy
- Information gain
- Gini impurity
- Tree depth
- Pruning
- Overfitting

## **18. Random Forest**

- Ensemble learning
- Bagging
- Bootstrap sampling
- Random feature selection
- Feature importance

## **19. Support Vector Machines**

- Hyperplanes
- Margins
- Support vectors
- Linear SVM
- Nonlinear SVM
- Kernel concept

## **20. Naïve Bayes**

- Bayes theorem
- Conditional probability
- Independence assumption
- Gaussian Naïve Bayes
- Multinomial Naïve Bayes

# **LEVEL 7 — MODEL EVALUATION**

## **21. Regression Metrics**

- MAE
- MSE
- RMSE
- $R^2$
- Adjusted  $R^2$

## **22. Classification Metrics**

- Confusion matrix
- True Positive
- True Negative
- False Positive
- False Negative
- Accuracy
- Precision
- Recall
- F1-score
- Specificity
- ROC-AUC
- PR-AUC

## **23. Cross-Validation**

- K-Fold
- Stratified K-Fold
- Leave-One-Out
- Validation strategy

## **24. Bias & Variance**

- Bias
- Variance
- Bias-variance tradeoff
- Underfitting
- Overfitting
- Generalization

# **LEVEL 8 — UNSUPERVISED LEARNING**

## **25. K-Means Clustering**

- Clustering concept
- Centroids
- Distance
- Inertia
- Elbow method
- Choosing K

## **26. Hierarchical Clustering**

- Agglomerative clustering
- Dendrogram
- Linkage methods

## **27. DBSCAN**

- Density-based clustering
- Core points
- Border points
- Noise
- Epsilon
- MinPts

## **28. PCA**

- Dimensionality reduction
- Principal components
- Variance
- Projection
- Explained variance

# **LEVEL 9 — FEATURE ENGINEERING**

## **29. Feature Engineering**

- Feature creation
- Feature transformation
- Feature extraction
- Feature selection
- Interaction features
- Polynomial features
- Binning

## **30. Feature Selection**

- Filter methods
- Wrapper methods
- Embedded methods
- Correlation-based selection
- Recursive Feature Elimination

# **LEVEL 10 — OPTIMIZATION & TUNING**

## **31. Gradient Descent**

- Objective function
- Cost function
- Gradient
- Learning rate
- Batch gradient descent
- Stochastic gradient descent
- Mini-batch gradient descent

## **32. Hyperparameter Tuning**

- Parameters vs hyperparameters
- Grid Search
- Random Search
- Search space
- Bayesian optimization
- Optuna

# **LEVEL 11 — GENERALIZATION & MODEL ANALYSIS**

### **33. Data Leakage**

- Target leakage
- Train-test contamination
- Preprocessing leakage
- Preventing leakage

### **34. Class Imbalance**

- Imbalanced datasets
- Class weights
- Oversampling
- Undersampling
- SMOTE

### **35. Error Analysis**

- Prediction errors
- False positives
- False negatives
- Error distribution
- Identifying systematic errors

## **LEVEL 12 — PRACTICAL MACHINE LEARNING**

### **36. Scikit-learn**

- Dataset loading
- Preprocessing
- Model training
- Prediction
- Evaluation
- Cross-validation
- Hyperparameter tuning

### **37. ML Pipelines**

- Pipeline
- ColumnTransformer
- Preprocessing pipeline
- Training pipeline
- Reproducibility

### **38. Model Persistence**

- Saving models
- Loading models
- Joblib
- Pickle